Title: Analysing the Evolution of IETF Standardisation Using the Updated Datatracker Database

Abstract: Previous studies have analysed the evolution of RFC publications and the characteristics of the IETF using historical Datatracker data. However, the database has continued to evolve, providing more complete publication records and author information.

This project aims to investigate how the IETF has changed over time by analysing the latest Datatracker SQLite.including publication trends across technical areas, Working Group activity, development time from Internet-Drafts to RFCs, draft revision complexity and organisational participation.

Introduction: The Internet Engineering Task Force (IETF) is responsible for developing many of the Internet standards used today. These standards are published as Request for Comments (RFCs).
This project uses an updated IETF Datatracker database to study how the standardisation process has changed over time. The analysis focuses on RFC publication trends, Working Group activity, draft development time, revision numbers, and author affiliations.

Dataset

The analysis uses the latest Datatracker SQLite database provided by the IETF.

Two databases are available:

  ietfdata-dt.sqlite

  ietfdata-ma.sqlite

This work currently focuses on ietfdata-dt.sqlite, which contains information about

  RFCs

  Internet-Drafts

  publication history

document relationships

Working Groups

authors

affiliations

Compared with the dataset used in the original paper, the updated database contains RFC records until 2025.

Methodology

The analysis is implemented in Python using SQLite and pandas.

I explored some several document relationships.

RFC publication dates are obtained from **published_rfc** events.
Draft-to-RFC mappings are obtained through the **became_rfc** relationship.
Historical Area assignments are reconstructed using **group history** records.
Author affiliations are extracted from the **document-author table.**

Preliminary Results

Figure 1

This figure shows that RFC publication activity has changed over time.

Different ARES has changed over time.

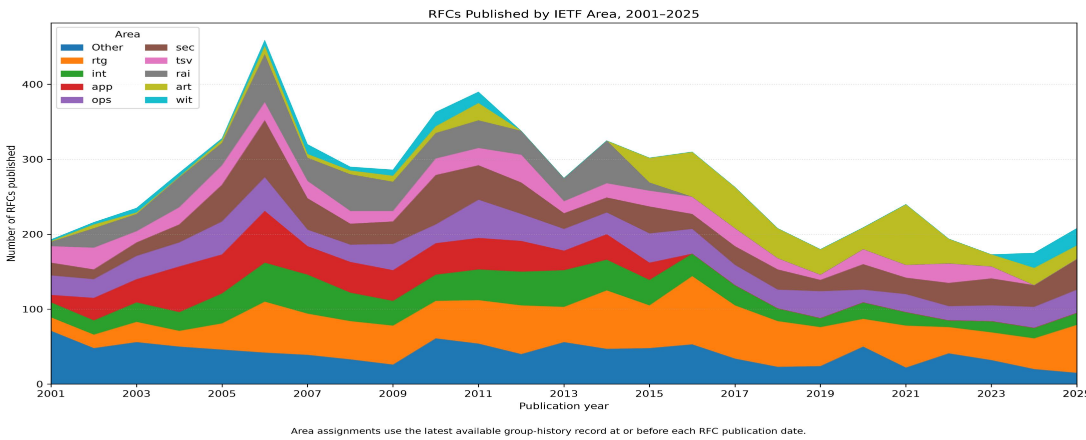
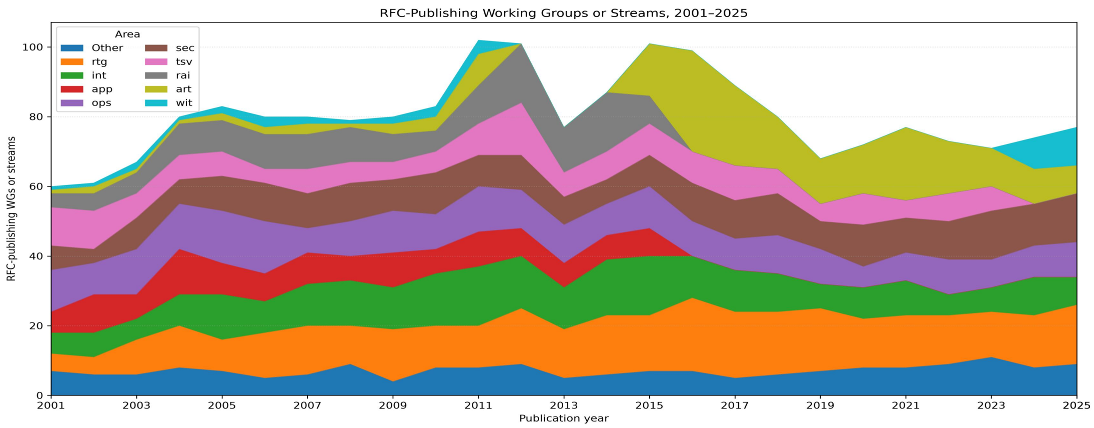


Figure 2

RFC-publishing Working Groups

The number of active RFC-publishing Working Groups fluctuates over time.

The results indicate that changes in RFC publication volume are associated with the creation and retirement of Working Groups.



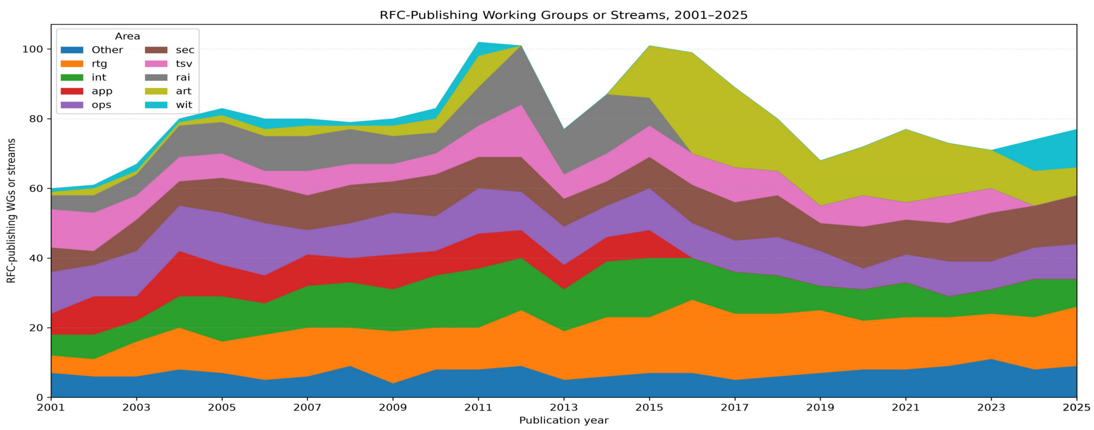
Each Working Group or other publication activity is counted once per year, regardless of the number of RFCs it published. Other includes IRTF research groups, legacy RFCs and non-IETF streams.

Figure 3

Days from first Internet-Draft to RFC publication

The results shows that the development time generally increases over recent years.

The overall trend suggests that standardisation requires longer development cycles than in previous years.



Each Working Group or other publication activity is counted once per year, regardless of the number of RFCs it published. Other includes IRTF research groups, legacy RFCs and non-IETF streams.

Figure 4

Internet-Draft revisions before publication

The median number of Internet-Draft revisions increases steadily.

This indicates that Internet standards generally require more revision cycles before publication, suggesting increasing review effort and technical complexity.

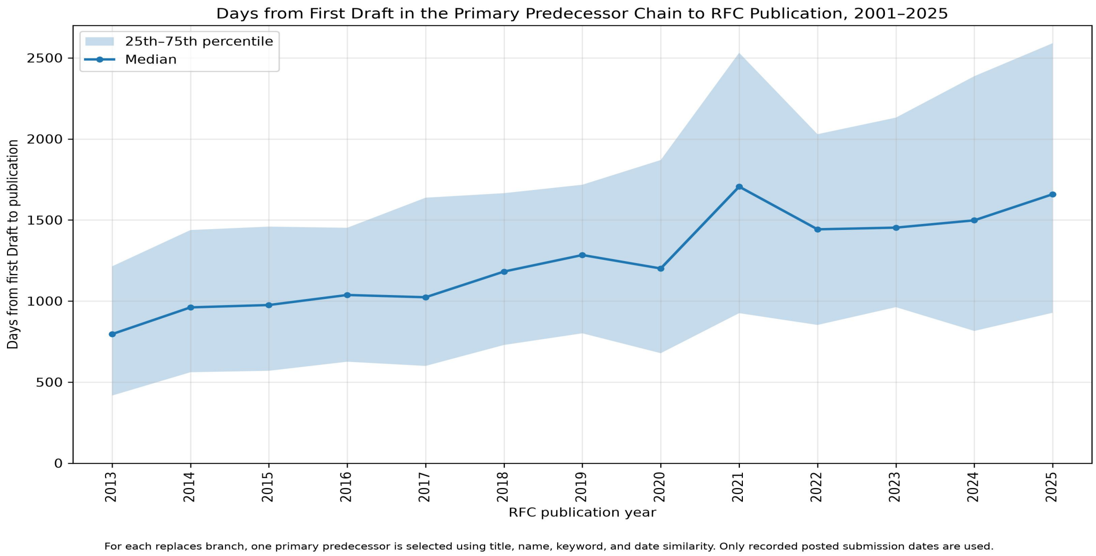


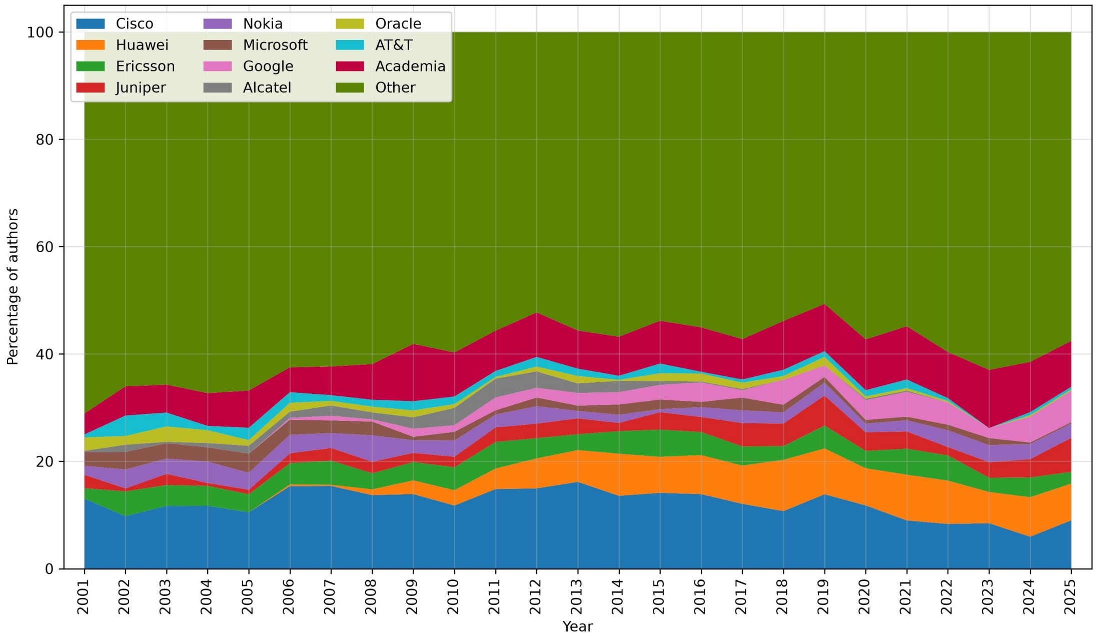
Figure 13

Author affiliations

Industry organisations continue to contribute the majority of RFC authors.

Cisco remains one of the largest contributors throughout the observation period.

Author affiliations were primarily obtained from the affiliation field in the ietfdata-dt.sqlite document-author records.



Discussion

The preliminary analysis suggests several observations.

First, the technical focus of the IETF has evolved alongside changes in Internet technologies.

Second, the increasing number of draft revisions and the longer development time indicate that the standardisation process has become more complex.

Finally, Several large technology companies remain consistently represented, but academic institutions account for a comparatively smaller proportion of recorded author affiliations.

Future Work

Future work will improve the accuracy of the data analysis and validate the current results using additional IETF records. I have to analyses the situation of Working group and author participation. Finally, the findings will be compared with previous studies to evaluate how the IETF has evolved over time.